

SAUDA, Norway

WMO: 014240

Lat: 59.649N

Lon: 6.363E

Elev: 20

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.8	(c) 14.3	(d) 2.2	(e) 6.2	(f) 15.8	(g) 5.4	(h) 7.3	(i) 18.8	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	78.7	60.8	74.3	59.9	70.7	58.2	63.8	73.7	62.1	70.5	60.3	67.7	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	78.0	66.6	58.5	73.1	64.9	56.8	68.7	63.3	29.1	73.4	27.8	71.1	26.6	67.8	70.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	10.7	85.9	4.4	5.5	7.6	89.9	5.0	93.1	2.5	96.2	-0.7	100.2
			9.7	66.7	4.1	2.6	6.8	68.5	4.4	70.0	2.1	71.5	-0.8	73.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.4	31.1	31.7	36.7	44.4	51.8	56.7	61.5	59.6	53.6	45.4	39.2	32.6	(6)
	DBStd	12.25	7.21	6.90	5.39	5.63	7.17	4.98	5.56	4.11	4.49	5.52	5.99	7.92	(7)
	HDD50	2804	587	512	412	186	62	4	0	0	17	159	325	540	(8)
	HDD65	7191	1052	932	877	618	414	255	141	175	341	607	774	1005	(9)
	CDD50	1141	0	0	1	18	118	205	356	298	126	17	2	0	(10)
	CDD65	51	0	0	0	0	5	6	32	8	0	0	0	0	(11)
	CDH74	405	0	0	0	0	60	65	241	39	0	0	0	0	(12)
	CDH80	81	0	0	0	0	13	4	61	3	0	0	0	0	(13)
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
(15) Precipitation	PrecAvg	95.0	11.8	9.6	8.4	5.1	4.1	4.3	5.1	6.9	7.9	9.8	10.2	11.3	(15)
	PrecMax	128.0	24.1	20.9	23.0	10.4	10.1	9.7	8.4	12.8	14.7	20.9	19.4	24.3	(16)
	PrecMin	60.6	0.8	0.3	0.4	1.7	0.2	0.6	0.9	1.7	1.7	1.7	1.1	0.6	(17)
	PrecStd	17.2	5.9	5.8	4.8	2.2	2.5	2.6	1.8	2.7	4.4	5.0	5.2	6.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.5	48.3	53.5	66.5	81.1	80.1	85.9	78.6	68.8	64.4	55.1	50.4	(19)
		MCWB	40.5	41.2	44.9	50.0	58.1	59.7	64.7	61.4	56.5	53.3	47.6	44.0	(20)
	2%	DB	44.2	45.1	48.6	63.1	73.9	75.6	80.6	73.4	66.3	57.5	52.3	47.4	(21)
		MCWB	39.9	39.4	40.9	48.8	56.7	58.4	63.5	60.0	55.3	52.4	44.7	42.4	(22)
	5%	DB	42.0	42.7	46.3	58.8	69.6	70.2	75.4	70.1	63.8	55.1	50.2	45.0	(23)
		MCWB	37.4	38.2	40.2	46.8	54.2	56.4	62.3	58.6	53.9	49.6	45.3	40.2	(24)
	10%	DB	40.0	40.3	44.4	54.4	65.0	66.3	71.1	67.2	61.4	53.0	47.9	42.8	(25)
		MCWB	35.7	37.1	39.6	44.5	52.6	54.6	60.1	57.8	53.9	47.9	44.6	38.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.9	43.5	45.4	52.0	60.7	62.3	67.1	65.0	59.8	56.7	49.9	46.5	(27)
		MCDB	44.5	46.4	50.7	62.4	76.3	74.3	78.8	73.6	63.2	59.2	52.9	48.7	(28)
	2%	WB	40.5	40.7	43.3	49.5	57.9	59.7	64.9	62.4	57.7	53.8	48.1	43.4	(29)
		MCDB	43.8	43.7	46.4	60.3	71.7	70.9	76.1	68.9	62.3	57.2	50.3	46.3	(30)
	5%	WB	38.3	38.9	41.6	47.7	55.3	57.6	63.3	60.8	56.4	51.2	46.1	40.9	(31)
		MCDB	41.1	41.7	45.0	56.8	67.5	67.4	72.8	66.6	60.8	54.3	48.3	44.3	(32)
	10%	WB	36.4	37.4	40.0	45.6	53.1	56.0	61.6	59.3	55.0	48.8	44.0	39.1	(33)
		MCDB	38.9	39.7	43.6	52.9	63.6	64.6	70.1	64.7	59.2	51.6	47.1	41.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.8	11.0	14.6	16.2	15.0	13.7	12.8	11.3	10.0	8.4	8.0	(35)
		MCDBR	10.8	10.4	12.6	22.0	24.5	23.0	21.9	18.0	15.5	11.8	10.6	10.5	(36)
	5% WB	MCWBR	7.4	6.7	8.0	11.4	10.7	10.5	8.8	8.0	8.0	7.5	7.3	7.7	(37)
		MCWBR	9.5	9.5	11.2	19.0	22.4	18.8	19.2	14.1	12.4	9.8	9.7	10.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.275	0.309	0.343	0.383	0.381	0.373	0.390	0.384	0.358	0.334	0.290	0.255	(40)
		taud	2.104	2.175	2.160	2.131	2.262	2.384	2.366	2.375	2.440	2.385	2.198	2.044	(41)
	Ebn at Noon		156	216	246	256	268	272	264	256	244	210	151	115	(42)
		Edh at Noon	15	24	33	40	38	35	34	32	25	20	13	10	(43)
(44) All-Sky Solar Radiation	RadAvg		99	292	695	1268	1505	1604	1445	1131	723	351	128	64	(44)
	RadStd		22	68	141	225	251	198	187	100	74	57	24	9	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page